

```

for all  $i, j \in [0, N) \times [0, N)$  do
  for  $k \in [0, d)$  do

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$$S_{i,j} := S_{i,j} + Q_{i,k}K_{j,k}$$

```

for all  $i \in [0, N)$  do
  for  $j \in [0, N)$  do

```

$$\tilde{P} = e^{S_{i,j}}$$

$$\begin{bmatrix} m_i \\ \ell_i \\ O_i \end{bmatrix} := \begin{bmatrix} m_i \\ \ell_i \\ O_i \end{bmatrix} \oplus_2 \begin{bmatrix} S_{i,j} \\ \tilde{P} \\ \tilde{P}V_j \end{bmatrix}$$

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for all  $i, j \in [0, N) \times [0, N)$  do
   $O_{i,j} := O_{i,j} / \ell_i$ 

```

decompose

```

for all  $i, j \in [0, N) \times [0, N)$  do
  for  $k \in [0, d)$  do

```

$$S_{i,j} := S_{i,j} + Q_{i,k}K_{j,k}$$

```

for all  $i \in [0, N)$  do
  for  $j \in [0, N)$  do

```

$$\tilde{m} = \max(m_i, S_{i,j})$$

$$\tilde{P} = e^{S_{i,j} - \tilde{m}}$$

$$\tilde{\alpha} = e^{m_i - \tilde{m}}$$

$$m_i := \tilde{m}$$

$$\ell_i := \alpha \ell_i + \tilde{P}$$

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for all  $k \in [0, d)$  do

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$$O_{i,k} := \alpha O_{i,k} + \tilde{P}V_{j,k}$$

```

for all  $i, j \in [0, N) \times [0, d)$  do
   $O_{i,j} := O_{i,j} / \ell_i$ 

```